

Introduction

- **Chronic wound infections** affect millions of people annually in the U.S. (Kot et al. 2020)
- Symptoms include increased pain and swelling at the wound site, pus, fever, fatigue, ect.
- These infections commonly involve **biofilms**, which are microcolonies of surface-attached bacterial cells which are highly resistant to antibiotics (Islam et al. 2022)
- Some researchers have resorted to more natural sources of inhibition, one of which is **honey**

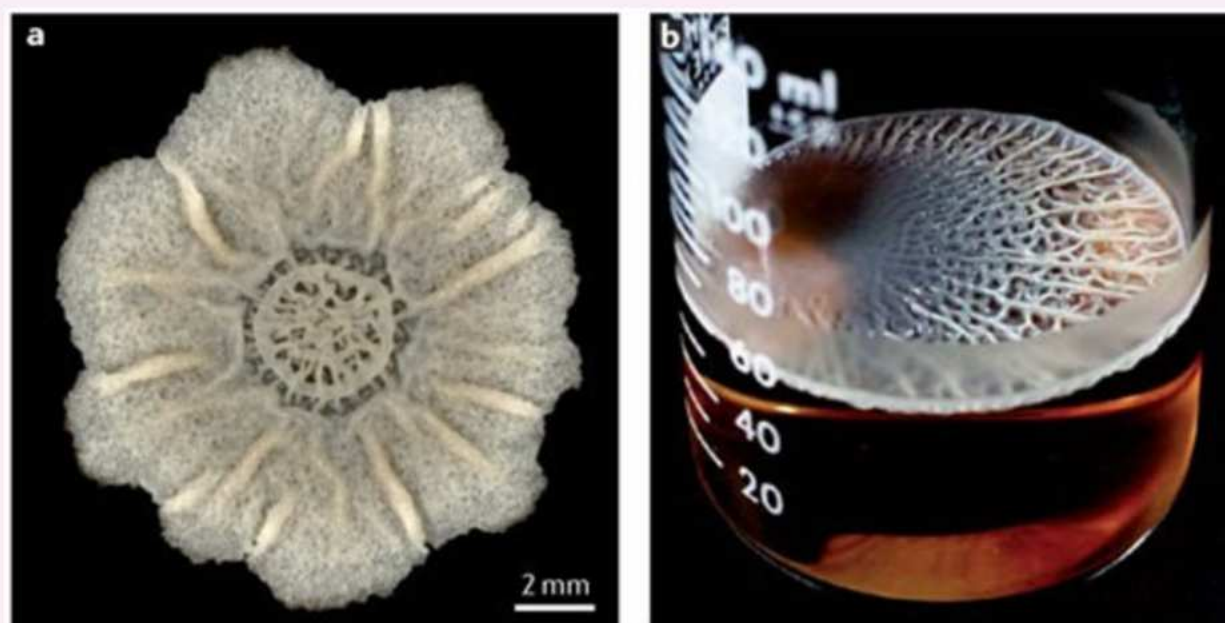
Manuka Honey Products



- Honey has been known to have healing properties for thousands of years
- Honey's key antibacterial component is **methylglyoxal**, or **MGO**
 - this promotes cell death due to its selective toxicity to bacterial cells (Neo et al. 2024)
- Monofloral **manuka honey**, has been shown to have larger amounts of total phenols (antioxidant compounds) and 20x higher MGO content than non-manuka types (Johnston et al. 2018).
- The phenolic compound **methyl syringate** is a **flavonoid** found in manuka honey, and is a marker substance commonly known for its strong anti-inflammatory properties
- This study aims to test the synergistic effects of MGO and methyl syringate on the inhibition of a *B.subtillis* biofilm

Research Question

Does the the phenolic compound methyl syringate enhance the antibiofilm properties of methylglyoxal in manuka honey through inhibition of a *Bacillus subtilis* biofilm?



Bacillus subtilis Biofilm

https://commons.wikimedia.org/wiki/File:Microscopic_image_of_Bacillus_subtilis_biofilm.jpg

Investigating the Synergy Between Methylglyoxal and Methyl Syringate Found in Manuka Honey Through Inhibition of a Bacillus Subtilis Biofilm

Hypothesis & Variables

Hypothesis: Higher concentrations of MGO manuka honey in synergy with methyl syringate will enhance antibiofilm effects due to methyl syringate being a marker substance in manuka honey.

IV: Concentrations of manka honey and methyl syringate used in synergy

DV; Inhibition of *B.subtilis* as measured by Crystal Violet assay (biomass) and Fractional Inhibitory Concentration (FIC) index

Controls: Untreated biofilm, killed biofilm, growth media, culture stock, manuka honey only, methyl syringate only, artificial honey solution only

Experimental Method

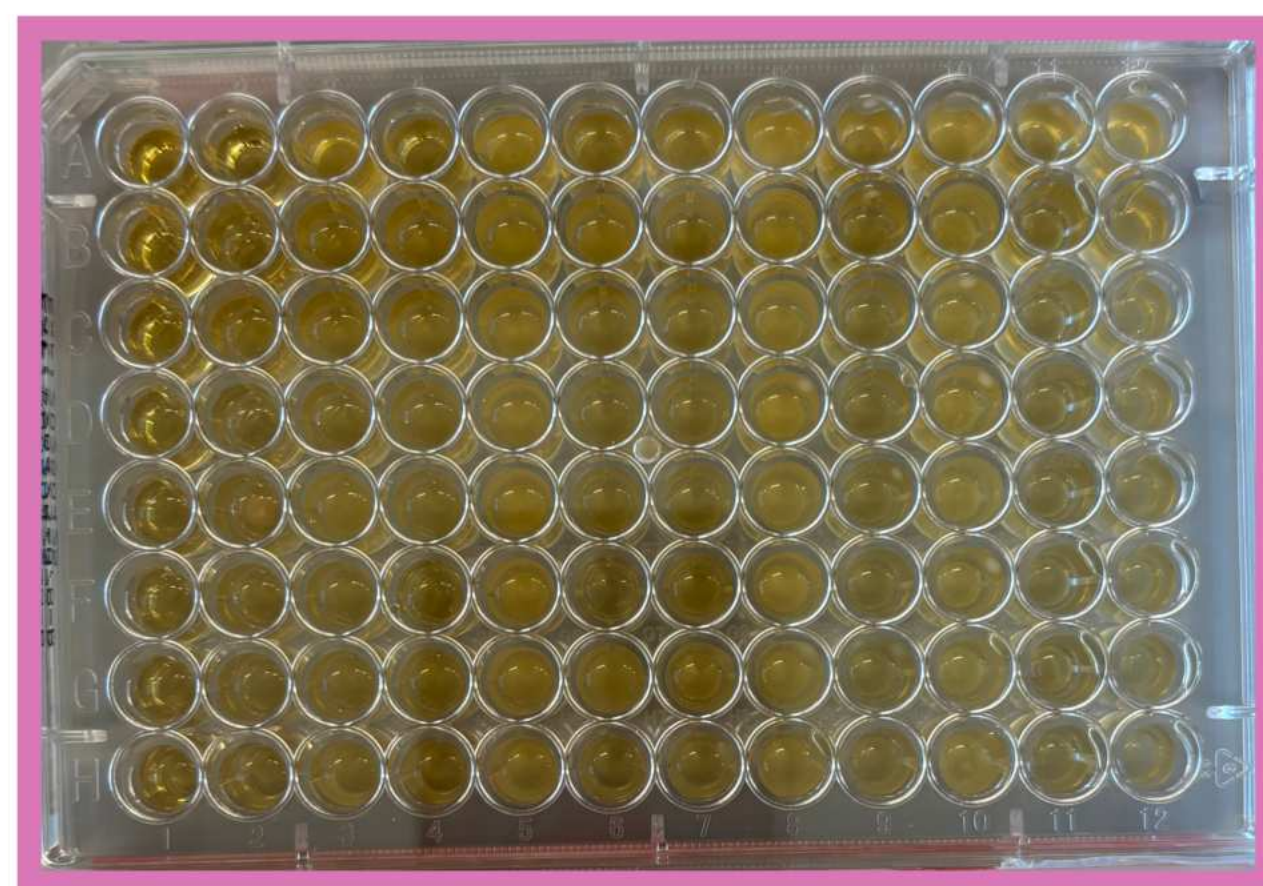
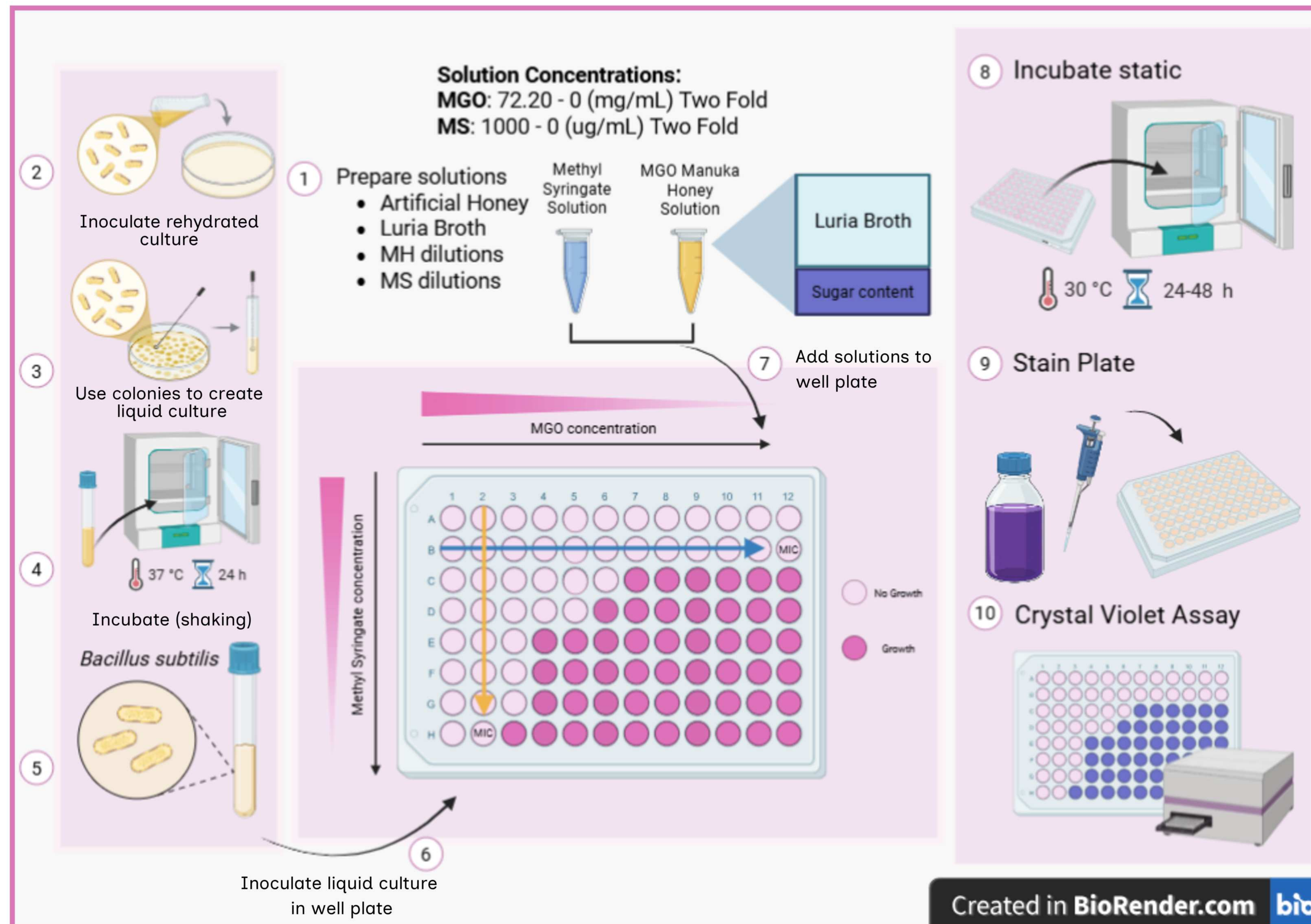


Plate 1: Before Staining
photo taken 03/24/2026

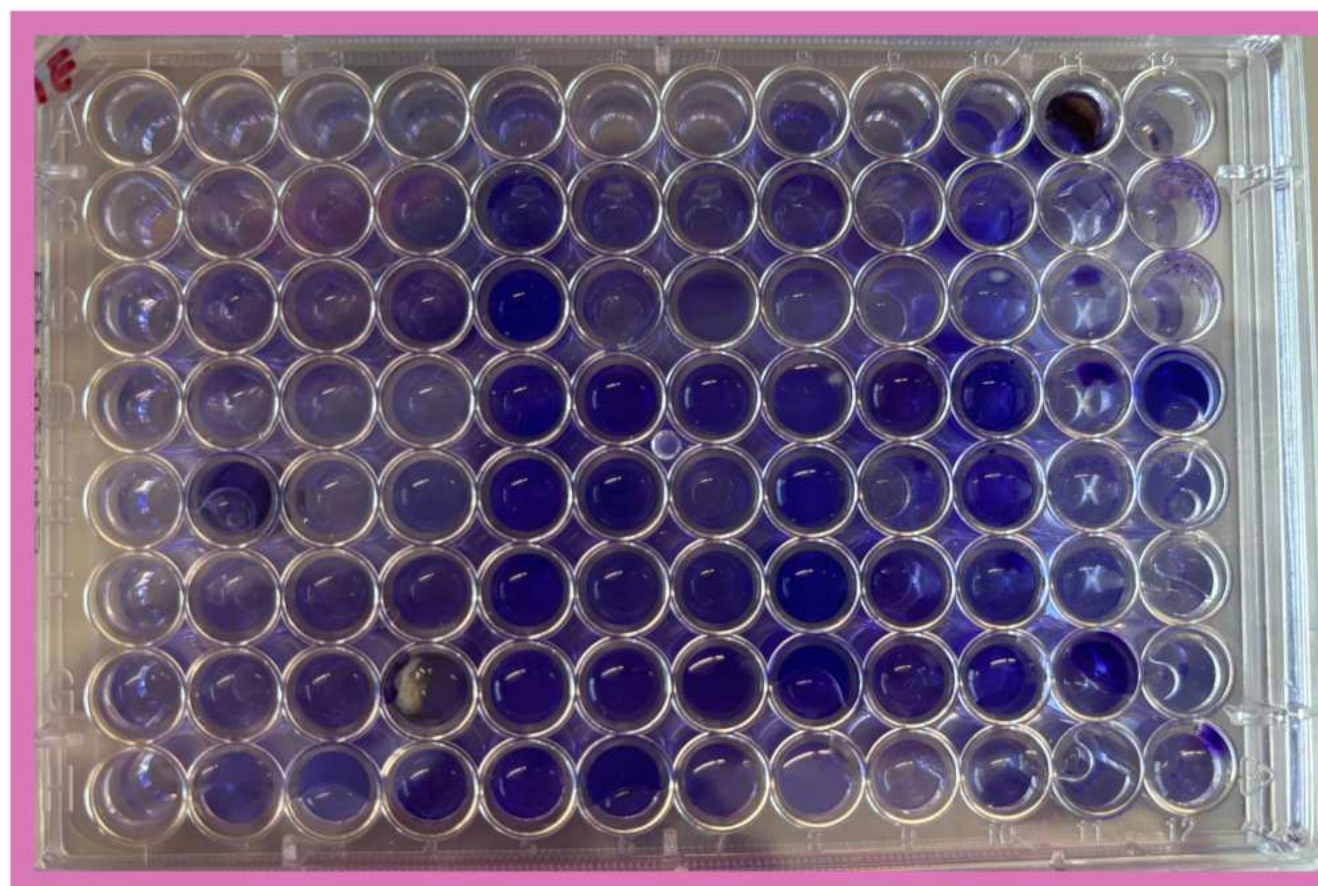


Plate 1: After Staining
photo taken 04/15/2026

Plate 1: Before and
After CV Assay

Data/Results

Table 1

Minimum Inhibitory Concentration for MGO and Methyl Syringate of Each Well

Plate #	Manuka MIC	MS MIC	FICI
1	18.05	1000	n/a
2	36.1	125	n/a
3	18.05	1000	n/a

Note. n/a values represent areas where data was unable to collected/
results were inconclusive

Discussion

Due to limited time constraints as well as many raodblocks faced, conclusive results were not reached for this study. However, the following results were observed:

- Biofilm inhibition was seen at **higher concentrations** of both compounds, which is consistent with past studies
- Control wells showed that there may have been an **immature biofilm** or the biofilm may not have grown properly
- FICI values were not calculated as the combined MICi values for MGO and methyl syringate were not observed after crystal violet staining
- Observed results indicate that **no synergy was present** between the experimental components

Conclusions & Future Work

- A clear conclusion was not drawn from this project
- However, past studies and the limited results which were observed indicate that **MGO and methyl syringate may have some synergistic effects against biofilms**
- A synergistic or additive effect was not seen as no FICI values were calculated, leading to **inconclusive results**.
- Future research for this experiment would include testing larger concentrations of MGO and **stronger biofilm masses**
 - Allow for clearer effects of inhibition after crystal violet staining
- Synergistic effects could be beneficial in aiding the many people affected by biofilm infections every year
- **Further research** on this topic is needed.

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- SDS INFO:
Methyl Syringate, Crystal Violet, Resazurin, PBS solution, Acetic Acid, B.subtilis